Aegis Station

Lunar Water Logistics

6 Million Tonnes of Lunar Water to Lunar Orbit, on a Fleet That Imports Nothing

Architecture Reference Document | Rev D — November 2026 | Not for Hardware Commitment

---

1. Executive Summary

Aegis Station is a rotating habitat in a 100 km lunar orbit, shielded by three metres of water around the outer hull of each of its three rings. At the 1g baseline that shield is 6.0 million tonnes. Lifting that mass from Earth is not a plan; it is a number that ends the program. Every tonne of shield water is therefore mined at the lunar south pole, carried to the pad in a standard 45-tonne cartridge, and flown to orbit by an autonomous tanker.

Rev B made one structural change to the chain. The tanker flies on liquid oxygen and liquid hydrogen split from lunar water at the pad, from its first flight. Rev A baselined methalox with methane shipped from Earth and treated hydrolox as a later upgrade. At fleet cadence that methane was a recurring import of hundreds of tonnes a day landed on the Moon, a supply chain larger than every lunar landing to date, repeated indefinitely. Anything a program has to keep buying from Earth is the thing that ends it. The import is designed out before the first flight, and the honest cost moves from the launch manifest to a refinery-class power plant on the surface.

Rev C corrects the shield mass to the full 3 m shell of the station's geometry, 6.0 million tonnes where Rev B carried 3.3, and resizes the campaign to hold the fill time. The architecture is otherwise unchanged and is stated here as a baseline, not a closed design: 82 tankers, 3,690 tonnes a day, a fill of roughly 4.5 years, and the Lunar Orbital Propellant Depot as the fleet's first customer before the station exists. Section 11 lists what does not yet close.

Rev D takes the vehicle cycle from stated to simulated. The fleet coordinator now flies 84 copies of the tanker's flight software on this document's schedule, a launch slot every 17.6 minutes and the pad steps of Section 5 at their stated times, and holds 82 launches a day on eleven pads with each pad occupied 2.6 hours. The same run shows where this document was loose: a berth that only swaps a cartridge is held about 1.3 hours, not 3.2; a tanker is busy about six hours of its day, so the fleet of 82 follows from one flight a day per airframe and not from the cycle; and the apron every tanker waits on has no stated way on or off it. No quantity in Section 9 changes. Section 5 carries the result, Section 11 the corrections and the new open items.

---

2. Water Requirement

The shield is a 3 m water layer on the outer hull of each ring, outboard of a 7 m utility band and a 40 m habitable depth. The requirement is set by geometry and the 1g rotation baseline, not by logistics.

Parameter	Value
Shield thickness	3 m of water, outer hull, all three rings
Shield radial span	47 m to 50 m from the ring centreline, outboard of the utility band
Shield volume, three rings	~6.0 million m^3
Water mass	~6.0 million tonnes
Olympic pool equivalent	~2,400
Source	Lunar south pole, permanently shadowed craters. Earth-sourced water is excluded in every phase.

Shield thickness is the multiplier on every number in this document: tankers, sorties, power, and years. The 3 m figure is held by the station design; the dose case for it is stated on the water-shield evaluation page, not here.

---

3. Delivery Architecture

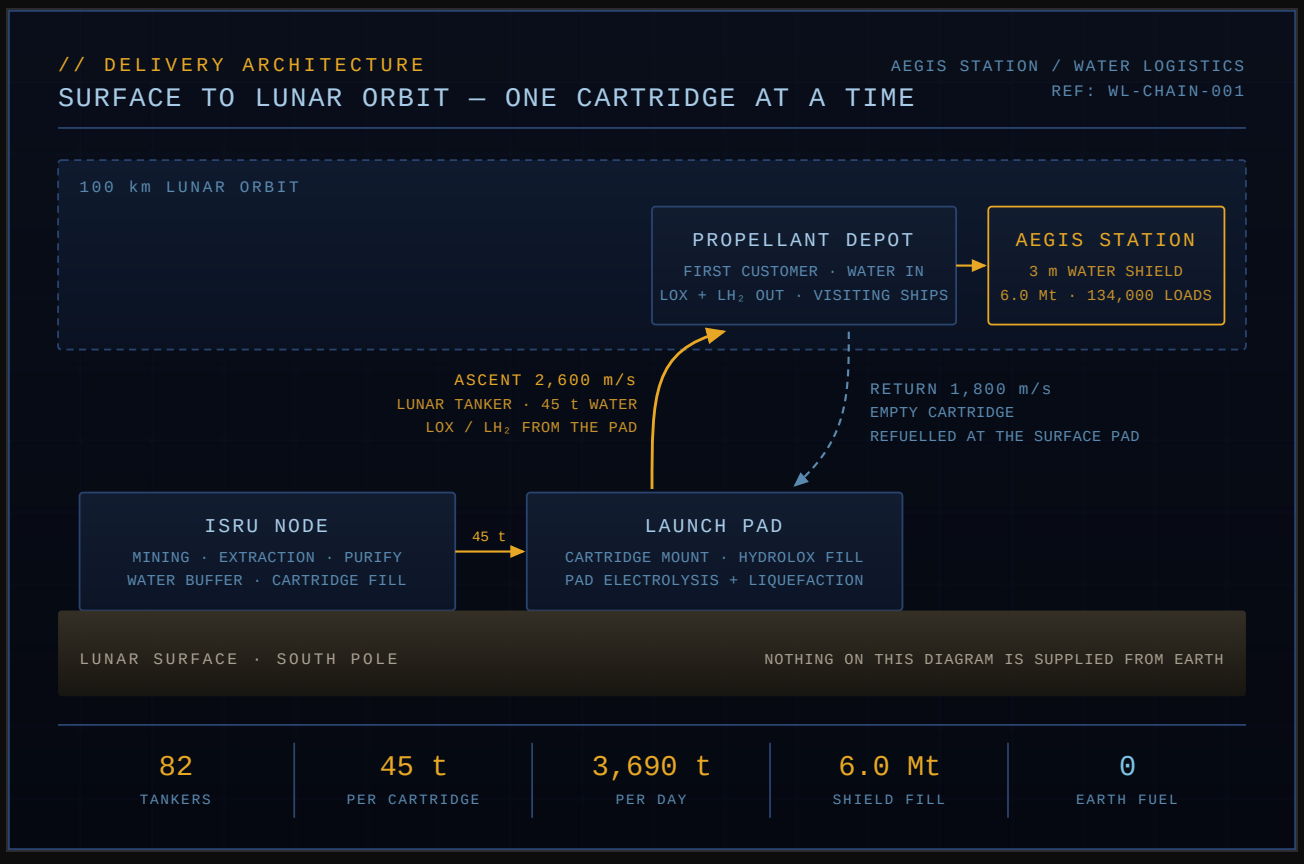


Fig. 1 — The delivery loop. Every sortie carries 45 t of water up and an empty cartridge down. Nothing on the loop is supplied from Earth.

Parameter	Value
Fleet	82 autonomous reusable tankers, VTOL, one ground console
Payload	45 t of water per sortie, in one Modular Tank Cartridge
Ascent	2,600 m/s to 100 km lunar orbit
Return	1,800 m/s, empty cartridge, round-trip propellant loaded at the surface pad
Daily throughput	~3,690 t/day
Fill time	~4.5 years at that throughput
Sorties, total	~134,000
Propellant	LOX / LH_2, both split from lunar water at the pad
Earth-supplied consumables	None, in any phase

The Modular Tank Cartridge

The cartridge is the atomic unit of the chain: a 10 m × 2.5 m structural cylinder holding a flexible bladder of ~49 m^3, filled by vacuum draw at the ISRU node and emptied by inert-gas displacement at the destination. It is cargo the tanker carries, not part of the tanker, and it has one water port and one gas port that are identical at the node, the pad, the depot, and the station. Water is never a cryogen anywhere in the chain. It needs no insulation, no vacuum-jacketed lines, and no boil-off management, which is why it is the commodity that moves.

Mission profile

0. **Depot first light.** A single tanker at low cadence lifts the first cartridges to the Lunar Orbital Propellant Depot and commissions end-to-end propellant production. Full fleet deployment is not required.
1. **Shield-fill campaign.** Sustained surface-to-orbit delivery to the station's shield reservoirs. This is the requirement that sizes the 82-vehicle fleet.
2. **Depot resupply.** Runs in parallel from first light and continues indefinitely after the fill. The depot is the fleet's first customer and a permanent one.
3. **Surface distribution.** Ballistic hops between ISRU and LUNET nodes at 160–940 m/s, redistributing water and LOX. Up to six hops per orbital propellant load.
4. **Rapid response.** One or two tankers held in standby; emergency water to a crewed outpost within 6–8 hours of dispatch.

---

4. Propellant: Hydrolox From Day One

The Moon has water ice. Electrolysis splits it into liquid oxygen and liquid hydrogen, the two highest-performing chemical propellants there are, and the only ones the Moon can make from a single feedstock without importing anything. The tanker burns both from its first flight. There is no methane

phase, no transition, and no import to taper off later. The Short Hopper already flies LOX/LH_2 for the same reason; since Rev B every vehicle on the surface burns the same ISRU product.

Parameter	Rev A (methalox)	Rev B (hydrolox)
Propellant	LOX / CH_4, methane imported from Earth	LOX / LH_2, both from lunar water
Specific impulse, vacuum	~360 s	~450 s
Launch mass	~102 t	~95 t
Ascent propellant	~52 t	~42 t
Round-trip propellant	~57 t	~45 t
Thrust, engines	~165 kN, 3 × 60 kN	~155 kN, 4 × 40 kN, BE-7 class, engine-out capable
Recurring Earth import per sortie	~12.5 t methane	None

Rev B propellant masses follow the rocket equation at the same payload, delta-v, and implied dry mass as Rev A. They are not a new sizing pass; see Section 11.

The hydrogen objection

Liquid hydrogen at 20 K is hard to store and harder to move. This architecture answers that the same way LUNET does: *produce near, consume fast.* A tanker fills from a liquefier a few metres from the pad and is empty in orbit within hours. That is the shortest hydrogen dwell any lunar vehicle will have. The long-dwell problems, boil-off, thermal-cycle fatigue on transfer lines, and embrittlement, belong to depots holding LH_2 for months, not to a vehicle that burns its load the same day it is filled. Zero boil-off storage at 20 K has been demonstrated at twice the previous state of the art, and the industry's own next-generation lunar lander is hydrolox for precisely this reason: it is the propellant ISRU can actually make.

The honest cost

Hydrogen is a ninth of water by mass, so a full tanker load means splitting several times its own hydrogen mass in water and liquefying the result. The tanker pad is therefore not a LUNET fill node, which is sized at tens to a few hundred kilowatts. It is a refinery-class facility with power to match, and it is the same plant the depot and the surface network already assume exists. That is one thing to build, once, on the Moon, rather than something to keep launching. Its size scales with sortie cadence and is an open item in Section 11.

> **Design rule.** No vehicle in the surface fleet may depend on a consumable that must be resupplied from Earth. *Methane failed this rule. Hydrolox passes it because the feedstock is the cargo.*

---

5. Refuelling: Method and Location

The loading-site capacity row in Section 11 makes pad turnaround the number the fleet keys off. Two decisions set it: how propellant gets aboard, and where. This section closes both under the fleet-analysis assumptions of Section 11: a full cartridge is at the pad, an empty is at the station, and propellant is available when called for.

What has to be moved each cycle

Item	Value
Round-trip propellant	~45 t: ~6.4 t LH_2 at 20 K, ~38.6 t LOX at 90 K
Tank volume	~90 m^3 LH_2 plus ~34 m^3 LOX, about 125 m^3, two and a half times the water cartridge
Fill rate, fluid transfer	Shuttle-era ground systems loaded LH_2 at ~10 kg/s. At 2–5 kg/s LH_2 and ~10 kg/s LOX, in parallel, a full load takes 60–75 minutes
Cryogenic connections per cycle	Two, either way. A fluid coupling connects a hose; a swappable module connects the module's feed to the engines. Neither option removes the connect cycle; they move it.

Method: fluid couplings versus swappable propellant modules

Turnaround step	Fluid couplings	Swappable modules
Land, safe, vent checks	15 min	15 min
Cartridge swap: unlatch empty, mount full, latch, check	50 min, in parallel with fill	50 min, in parallel with module swap only if two heavy handlers work one vehicle at once
Propellant aboard	Couple and chill lines 20 min, transfer 60–75 min, disconnect, leak check and purge 15 min	Disconnect and remove spent module 25 min, install, latch, connect and leak-check 35 min, chill engine feed 10 min
Inspection and prelaunch	30 min	30 min
Pad time, steps overlapped	**~2.6 h**	**~2.0 h**
Pad time, sequential	~3.4 h	~2.7 h

Consequence	Fluid couplings	Swappable modules
Pads for 82 sorties/day	9 at 2.6 h; 11 hold the cadence at the 3.2-hour ceiling	7 at 2.0 h; 11 allow up to 3.2 h
Vehicle dry mass	Tanks are primary structure, as on every lander flown	Tanks become cargo, so the airframe needs its own thrust structure plus a flight-rated structural, fluid and electrical interface: about +2–3 t

Consequence	Fluid couplings	Swappable modules
Payload per sortie	45 t	~42–43 t at the same launch mass, roughly 5 % more sorties over the campaign, so 5 % more engines, propellant, pad-hours and years
Where the cycle wear lives	Ground-side hose couplings: cheap, replaceable on a schedule, no flight mass. This is the LUNET rule for cryogenic lines applied to the pad.	A flight-loaded interface on the vehicle that must survive ~1,600 connect cycles and every launch and landing between them
Ground hardware	Buffered LOX and LH_2 store, transfer lines, one set of couplings per pad	The same store and lines, plus a fill bay, a module handling system for a ~50 t cryogenic unit, and a pool of ~45 modules at a few tonnes each
Boil-off	Fill on demand; losses are line chilldown per cycle	Filled modules wait on the apron and lose LH_2 while they do
Off-nominal	A failed coupling closes one pad	A failed module is one module; a failed interface grounds the vehicle

The module buys about 0.6 hours of pad time per sortie, which at fleet cadence is about two pads. Pads are among the cheapest things on the site. It pays for that with two to three tonnes of dry mass on every flight for the whole campaign, a heavy cryogenic handling system that the water cartridge does not need because water is not cryogenic, and a flight-rated interface that carries the same connect-cycle count as the hose coupling it replaces but under launch loads. The water cartridge is swappable because water tolerates it. Propellant does not get the same treatment for the same reason the network moves water and not hydrogen.

Location: surface only versus depot top-off

Consideration	Surface only	Surface plus depot top-off for the return leg
Return-leg propellant	Lifted from the surface, ~3–5 t	Taken on at the depot, ~3–5 t of hydrolox per sortie
Ascent saving	None	Not lifting the return propellant saves ~6–9 t of ascent propellant, worth roughly +3–4 t of payload per sortie
What it costs in orbit	Nothing	3–5 t of hydrolox per sortie is ~6 t of water electrolysed at the depot per sortie, ~530 t/day at fleet cadence, of order 160 MW of orbital power and a large 20 K storage bank
Berth time	Cartridge swap only	Cartridge swap plus a second cryogenic connection and transfer per cycle, roughly doubling a berth visit that is otherwise about 1.3 hours

Consideration	Surface only	Surface plus depot top-off for the return leg
Where the power already is	The refinery is on the surface and is sized for the ascent load anyway; the return leg is a ~10 % increment on it	Moves a tenth of the refinery into orbit for a tenth more payload

Depot top-off is the right answer for the Hauler, which cannot lift its return propellant from Earth economically, and for any vehicle that arrives from elsewhere. For a tanker that departs from a pad with a refinery on it, the return leg is a small increment on a plant that already exists, and buying it in orbit instead costs an orbital power plant and a second cryogenic connection on every cycle. Berth time is not what decides it: the simulated cycle below leaves the berths half idle. The tanker fills at the surface. Depot top-off stays available as an option for contingency and for post-fill operations when the depot has surplus, and the depot interface is unchanged.

> **Decision.** Fluid couplings at the surface pad, filled on demand from a buffered store, cartridge swap in parallel with propellant transfer, ground couplings treated as scheduled consumables. Turnaround target 2.6 hours, ceiling 3.2. *Swappable propellant modules and routine depot top-off are rejected for the water tanker. Fifteen pads are recommended over eleven for margin: eleven hold the cadence only at the 3.2-hour ceiling, fifteen hold it with a pad occupancy up to 4.4 hours, so a slow transfer or a pad out of service does not stall the fill.*

The cycle as simulated

The fleet coordinator runs this cycle closed-loop against the tanker's flight software: 84 instances, 82 flying and a standby reserve of two, eleven pads and eleven berths. Launch slots stand every 17.6 minutes. A tanker waits in a pad's line on the apron and is called onto the pad one servicing time before its slot, so a pad is held for servicing only. Each step ends when its time has passed and the vehicle's telemetry shows what the step was for. The pad steps and times are the fluid-coupling column above, with the transfer at 75 minutes. Over 2.14 simulated days:

Measure	Simulated	This document
Launches in 24 hours	82, in consecutive slots, none missed	82
Interval between launches	17.6 min mean, 17.0 to 18.1	One every 18 minutes
Pad held, call to liftoff	2.58 h	2.6 h target, 3.2 h ceiling
Pads in use	8.9 of 11 on average, never more than 9	9 at 2.6 h
Berth held, capture to release	1.34 h	3.2 h ceiling; no steps stated
Berths in use	4.6 of 11 on average, never more than 5	11
Ships queued for a berth	Never more than 1	Not stated
One tanker, liftoff to next liftoff	23.9 h	One flight a day

The pad arithmetic of this section holds: 82 a day on nine pads at 2.6 hours, with two of the eleven idle at any moment. What the run adds to the steps above is its own, not this document's, and is assumed until hardware says otherwise: the 50-minute swap divided into unlatch 5, pull 15, seat 15, latch 5 and

check 10 minutes; and a berth visit of 15 minutes of safing, the same 50-minute swap seating an empty cartridge, and 15 minutes of departure checks. The transfer at 75 minutes for 45 t is 8.3 kg/s of oxygen and 1.4 kg/s of hydrogen, so oxygen sets the time and the hydrogen rate sits under the 2–5 kg/s quoted above.

What the run does not show: every step took exactly its stated time, so the 0.6 hours between the 2.6-hour target and the 3.2-hour ceiling was never drawn on, and the case for fifteen pads, a slow transfer or a pad out of service, is still argued and not measured. Assumptions A1 to A3 of Section 11 are assumed in the simulation exactly as here. Nothing in it waits for a pad to come under the orbit.

---

6. The Depot as First Customer

Every lunar-water architecture faces the same deadlock: surface extraction is hard to justify without an orbital customer, and orbital hardware is hard to justify without a proven resource. The Lunar Orbital Propellant Depot breaks it by being the earliest useful thing in lunar orbit. Its hardware can be emplaced and stand dormant; its commissioning date is set by the first cartridge to arrive, and its existence is what makes that date matter.

- **Water in, propellant out.** Depot electrolysis produces LOX and LH_2 from cartridge water. With a hydrolox fleet the hydrogen is a product, not a vented byproduct: the depot holds both propellants for visiting vehicles and for contingency top-off of the tanker fleet. Routine tanker refuelling is at the surface pad, per the decision above, so the depot is not in the daily cycle.
- **Methane for the Hauler only.** The Earth–Aegis Long-Hauler still burns methalox for its return; it brings its own methane as manifest cargo. That import is declared, bounded, and never touches the tanker fleet.
- **Standard interfaces.** The cartridge and its two ports are the same at the depot as at the pad. A depot that only its builder can dock with is a fuel tank, not infrastructure.
- **Beachhead.** The depot's pressurised node is the program's first shirt-sleeve presence in lunar orbit, before any ring is assembled.

---

7. ISRU and Extraction

Ice-bearing regolith is excavated from permanently shadowed craters near the south pole and heated by microwave or radiant thermal rigs; the ice sublimes, is cold-trapped, condensed, and filtered to a single water grade that serves both cargo and propellant. An ambient surface buffer absorbs production swings and feeds two consumers: cartridge fill for the tanker, and pad-side electrolysis and liquefaction for the tanker's own propellant.

- Deposit assessment precedes extraction; the subsurface ice assessment capability and the WOK prospecting rover are the instruments aimed at that gate.
- Mining, extraction, and transfer are autonomous and modular; power is polar solar with fission for the lunar night.
- For every tonne delivered to orbit, roughly another tonne and a quarter is split for propellant. The mining rate, not the delivery rate, sizes the node.

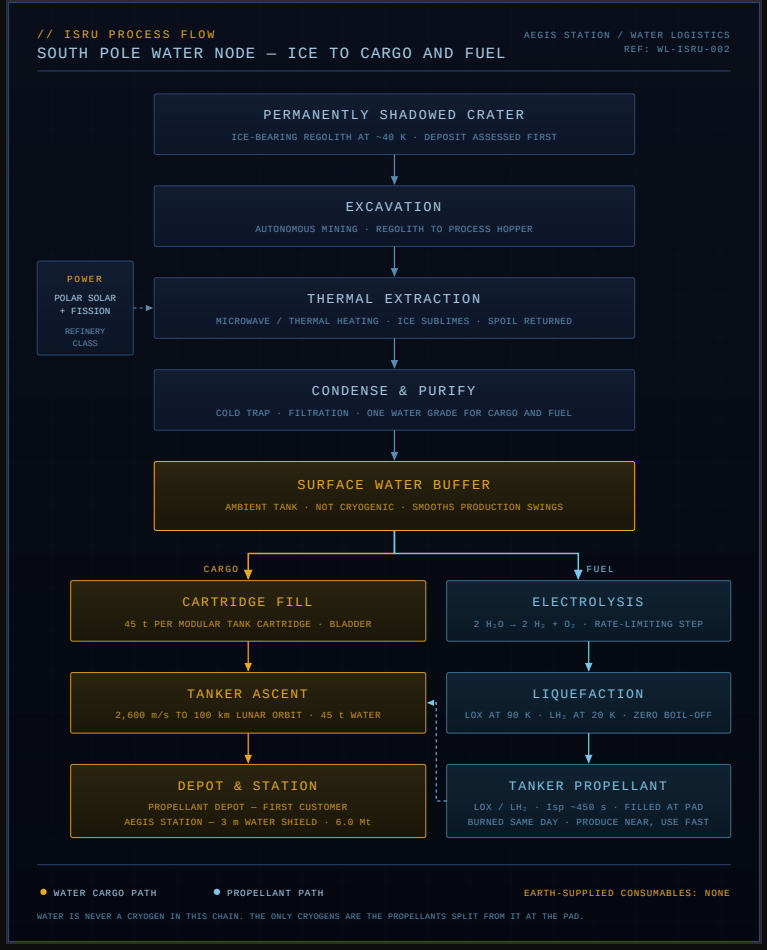


Fig. 2 — From ice in the crater to water in a cartridge and hydrolox in the tanker. The buffer is ambient; the only cryogens are the propellants split from it.

---

8. Logistics Infrastructure

Surface

- Launch pads with dust control and blast deflection
- An apron beside the pads where tankers wait between flights, about sixty of the 82 at any moment; how a tanker moves between apron and pad is open (Section 11)
- Cartridge fill, staging, and ground inventory of displaced inert gas
- Pad-side electrolysis and liquefaction: the refinery that fuels the fleet
- LUNET utility nodes for surface mobility, refuelling, and inter-node exchange
- Fleet Commander ground console: dispatch, pad deconfliction, berth queuing, standby reserve

Orbital

- Lunar Orbital Propellant Depot: cartridge reception, electrolysis, LOX and LH_2 storage, tanker top-off
- Station cartridge docking ports and water transfer lines to the shield reservoirs
- Autonomous transfer monitoring, inventory reconciliation, and fault management

Shield fill

Rings receive shield water from the start of assembly through evenly spaced fill ports. Fill sequencing preserves mass balance across each ring, and spin-up to 1g follows structural and systems verification of the complete ring assembly, so the shield is filled and balanced before the ring rotates.

---

9. Quantities

Cost is not yet estimated. This revision states what the fill moves and what moves it, which are the numbers a cost would be built from.

Item	Quantity
Shield water delivered	~6.0 million tonnes
Cartridge loads	~134,000
Water mined, cargo plus propellant	~14 million tonnes; ~8,300 t/day at fleet cadence
Tankers	82, each flying about 1,600 cycles
Pads	11 at the 3.2-hour ceiling, 9 in use at the 2.6-hour target; 15 recommended
Berths	11 carried; about 5 in use if a berth visit is a cartridge swap only (Section 11)
Engines over the campaign	~5,400–10,700; 800–1,600 t landed on the Moon
Surface refinery	Of order 1.5 GW continuous
Cartridge pool, mining, and depot hardware	Not yet sized

---

10. Long-Term Applications and Strategic Value

The extraction and delivery chain does not retire when the shield is full. The same fleet, pad, and depot serve depot resupply, surface distribution, emergency response, and propellant staging for anything leaving cislunar space, still without an Earth-supplied consumable.

- LOX and LH_2 production for cislunar and deep-space vehicles
- Depot refuelling in lunar orbit and at Earth–Moon staging points
- Life support reserves and shielding water for future habitats and platforms
- Water as a priced, deliverable orbital commodity with published interfaces
- A replicable model for any station built where water is available

The water economy doesn't end with Aegis. It begins with it.

11. Open Items

Rev B fixed the propellant decision. Rev C resized the campaign for the corrected shield mass and left the vehicle cycle as stated. Rev D has the cycle simulated (Section 5) and revises the rows below against it. The following do not yet close and are carried as open, not hidden.

Scope and assumptions for the fleet analysis

The tanker fleet is analysed as a vehicle cycle between a surface pad and a station berth. Three things are assumed rather than analysed here, because each belongs to another part of the program:

A1 **A full cartridge is available at the pad whenever a tanker is ready to lift.** Mining, extraction, cartridge fill, and the size of the cartridge pool are upstream of the fleet and sized separately.

A2 **An empty cartridge is available at the station whenever a tanker berths.** Shield-side discharge and cartridge handling are station systems.

A3 **Refuelling is timely wherever it happens, surface or station.** Refinery and depot capacity are their own items; they set the ceiling on cadence but are not part of the vehicle cycle.

Under those assumptions the fleet's own limits are pad-hours, berth-hours, orbital phasing, airframe availability, and turnaround mechanics. Where refuelling happens and how propellant gets aboard are decided in Section 5; the turnaround target that follows from it is carried below.

Item	Status
Daily turnaround	3,690 t/day at 45 t per sortie is one flight per tanker per day for the whole campaign, about 1,600 cycles per airframe. This is a requirement, not an assumption; the fill time is derived from it. It is an airliner cadence in an environment gentler than any reusable booster faces: no reentry, no aerodynamic load, one-sixth-g landings. What it demands is stated in the rows below: engines as scheduled replacement items, a pad turnaround of 2.6 hours, and a refinery sized to the cadence. No reusable orbital-class vehicle has yet flown daily; that is what the fleet has to demonstrate, not a reason to plan for less. **Rev D:** the cycle itself is about six hours, 2.6 at the pad, about two in flight and 1.3 at the berth, so a tanker flying once a day waits on the apron about eighteen hours of the twenty-four and about sixty of the 82 are parked at any moment. Throughput is bounded by pad-hours, about 100 launches a day on eleven pads at 2.6 hours, not by the number of tankers. The fleet of 82 is therefore what one flight a day per airframe requires. Flying each airframe two or three times a day would deliver the same water with a third to a half of the airframes and the same number of engines, sorties and pad-hours; whether airframe life or airframe count is the cheaper thing to spend is not decided here, and 82 is carried.
Engine pipeline	Each engine position sees roughly 500 hours of burn over an airframe life, many times any demonstrated engine life, so engines are scheduled replacement items. At 50–100 sorties per engine set, four engines per vehicle, the campaign consumes about 5,400–10,700 engines, three to seven a day from a single printed part number, 800–1,600 t landed on the Moon. Engine life is the lever: 200 sorties per set quarters the count. Life-expired engines stay at the pad as the first refurbishment demand on the surface.

Item	Status
Vehicle dry mass	The vehicle must close at 45 t of payload to lunar orbit. The stated launch and propellant figures imply roughly 5 t of structure, tanks, engines, and cartridge shell, which is aggressive for a lander. A bottom-up mass budget that closes at 45 t, with the launch mass that requires, is the open item. Not done in this revision. **Rev D:** the vehicle the flight software flies is not this one. It is 12 t dry with three 120 kN engines and 102 t at liftoff, budgets 1,870 m/s for each leg with 3 % and 10 % reserves where Section 3 states 2,600 and 1,800 m/s, and closes the round trip on the same 45 t of propellant, reaching orbit with 9.3 t and landing with 1.2 t. Each vehicle closes on its own figures; they have not been reconciled. The flight-profile work also finds a liftoff thrust-to-weight of two or better is needed to leave the pad, about 330 kN at that mass, against the four small engines counted in the row above.
Payload per sortie	**New in Rev D.** This document counts 45 t of water per sortie. The cartridge the flight software carries is 45 t full with a 1 t shell that comes back, 44 t of water delivered: 3,608 t/day at 82 sorties, 2 % under the 3,690 the fill time is derived from. Either the full cartridge is 46 t or the fill is about five weeks longer. Not resolved.
Surface power	Electrolysis and hydrogen liquefaction at fleet cadence are of order 1.5 GW continuous, about 18 MW per daily sortie. LUNET node sizing does not cover this; the pad refinery is a separate facility and is not yet sized.
Cartridge reuse	The cartridge is infrastructure, not a consumable, but its cycle life and bladder material are undefined. Fleet size of cartridges bounds throughput.
Loading-site capacity	Eighty-two sorties a day is one launch every 18 minutes. With eleven pads and eleven berths, each tanker may hold a pad for at most 3.2 hours and a berth for at most 3.2 hours, all steps included. The flight software's dispatch ADR carried a 4–8 hour placeholder for the swap; this requirement superseded it and the fleet coordinator now runs to it (Section 5). Pads are cheap against megawatts: seventeen pads relax the turnaround to five hours. A tanker waiting on propellant or a cartridge must not hold a pad; it waits on an apron, so pad occupancy measures servicing only. **Rev D:** the pads behave as stated, nine in use at 2.6 hours. The berths do not: earlier revisions called the berth as tight as the pad and gave it the same ceiling without any steps. A berth visit that is a cartridge swap and checks is about 1.3 hours and keeps five berths in use. Eleven are carried until the berth's steps are stated by the station side; shield-side discharge or cartridge handling done while the tanker is berthed would lengthen the visit, and nothing here says whether it is.
Apron and landing	**New in Rev D.** Section 5 starts the pad clock at landing and ends it at liftoff, 2.6 hours. A tanker flying once a day is on the ground about twenty hours, so it does not stay on the pad it landed on, and this document has said only that it waits on an apron. Either a tanker lands on a pad and is moved off and back, which is two moves of an empty tanker and a landing that occupies a pad outside the 2.6 hours; or it lands on the apron, which then needs the dust control and blast deflection the pads have. The simulation takes the second without the cost: a tanker lands into a pad's line, reaches the pad in no time, and has its safe-and-vent checks when it is called. The apron's size, its surface, and how a tanker moves across it are not designed.
Orbital phasing	**New in Rev D as a row;** named among the fleet's limits since Rev B and never analysed. A tanker leaving a berth can reach a given pad only when the pad is near the orbit's ground track, and a launch reaches the station only in a window. Neither the 17.6-minute slots nor the berth release in the simulation account for it. The effect on berth time and on the evenness of the launch cadence is open.

Item	Status
Turnaround target	Closed to a target by Section 5: fluid couplings at the surface, 2.6 hours with cartridge swap and propellant transfer overlapped, 3.2 hour ceiling for eleven pads. Still open: whether a 2.6 hour turnaround is achievable in practice with the cargo and coupling hardware as designed. Fifteen pads recommended for margin. **Rev D:** simulated at the stated step times the target holds, 2.58 hours a pad with every launch in its slot. That shows the steps add up and the schedule closes; it does not show the steps can be done in those times, and no step was allowed to run long. The margin case, step times that vary and a pad out of service on eleven pads and on fifteen, is the next run.
Airframe availability	Eighty-two sorties a day requires eighty-two airframes available every day. Airframes in engine swap or inspection are in addition to that number, not subtracted from it. The size of the maintenance rotation follows from engine-swap time and inspection intervals and is not yet set.
Resource base	Shield water plus propellant water is about 14 million tonnes against polar ice estimates that span two orders of magnitude. Deposit assessment is the gate, and it is why the program is waiting.

---

12. Changes From the Previous Revision

Rev C to Rev D

Item	Rev C	Rev D
Vehicle cycle	Stated	Simulated closed-loop against the flight software at the stated step times: 82 launches a day held on eleven pads (Section 5)
Pad time, sequential	~3.5 h	~3.4 h, the sum of the steps with a 75-minute transfer
Berth time	3.2-hour ceiling, "as tight as the pad"	About 1.3 hours for a cartridge swap; five of eleven berths in use; eleven carried pending the station's berth steps
Case against depot top-off	Berth time and an orbital power plant	An orbital power plant and a second cryogenic connection per cycle; berth time is not scarce
Fleet size	82, sized by the fill	82 carried; stated as following from one flight a day per airframe, the cycle being about six hours and pads the bound on throughput
Apron	Named	Listed as infrastructure; landing place and movement to the pad opened as an item
Vehicle	One vehicle, ~5 t dry, open	The flight software's 12 t vehicle recorded beside it, unreconciled
Payload per sortie	45 t of water	45 t carried; the 1 t cartridge shell, 44 t delivered, opened as an item
Orbital phasing	Named as a limit	Opened as an item
Quantities		Unchanged, with pads and berths now on separate rows

Rev B to Rev C

Item	Rev B	Rev C
Shield water	~3.3 Mt	~6.0 Mt, the full 3 m shell between 47 m and 50 m on the 350 m ring centreline
Olympic pool equivalent	~1,320	~2,400
Fleet	45 tankers	82 tankers, sized to hold the ~4.5-year fill
Daily throughput	~2,025 t/day	~3,690 t/day
Sorties, total	~73,333	~134,000
Launch cadence	One every 32 minutes	One every 18 minutes
Pads and berths	Six at the 3.2-hour ceiling, eight recommended	Eleven at the 3.2-hour ceiling, fifteen recommended
Engines over the campaign	~3,000–6,000	~5,400–10,700
Surface refinery power	Hundreds of megawatts	Of order 1.5 GW
Cost figures	Per-kilogram rates and campaign totals	Removed; Section 9 states quantities

Rev A to Rev B

Item	Rev A	Rev B
Tanker propellant	LOX/CH_4, methane imported from Earth, hydrolox as a later phase	LOX/LH_2 from lunar water, first flight onward
Earth-supplied consumables	~12.5 t methane per sortie for the first years	None
Water handling	"Cryogenic transport tanks", "cryogenic storage tanks", "cryogenic transfer lines"	Ambient water in bladder cartridges; only the propellants are cryogenic
Surface nodes	Remote Operations Nodes	LUNET utility nodes plus a refinery-class pad
Depot	Not described	First customer; commissions on first cartridge; hydrogen retained for tanker top-off
Open items	Not stated	Section 11
Refuelling	Not stated	Fluid couplings at the surface pad, fill-on-demand from a buffered store; swappable propellant modules and routine depot top-off rejected (Section 5)